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Planar-cellular-dendritic transformation in the fusion zone of a GH909 superalloy weldment



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ABSTRACT

Solidification behaviors and resultant crystalline growth of primary gamma (γ) and γ /Laves phases were systematically investigated in a GH909 superalloy laser weldment. Results showed that horizontal microstructural evolution varied along the thickness direction under Marangoni force and eruption of metal vapor from keyhole. Overall, there existed three distinct microstructures: γ trunks, eutectic γ and Laves distinguished by composition and band contrast map. Shoji-Nishiyama orientation relation was verified between γ and Laves in the eutectic constituents. For γ trunks in mid-penetration, its morphology displayed a planar-cellular-dendritic growth sequence. Texturally, there displayed an increasingly concentrated tendency of [100]-crystal direction along RD and high Schmidt factor was observed in planar and cellular zones. By recording and analyzing practical thermal field, an increasing solidification rate and a decrease of thermal gradient not only produced the planar-cellular-dendritic growth mode, but also increased γ primary spacing and grain size from planar zone to the dendritic. As for γ /Laves constituents, reduced eutectic reaction time resulted in a decreased fraction from planar growth to cellular zone. Tensile test and fracture analysis indicated that high Schmidt factor, relatively large grain size and high fraction of γ /Laves constituents jointly led to the failure in planar and cellular zones of the GH909 laser weldment.

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1. Introduction

High-strength, low-expansion Nb-bearing superalloys are primarily designed for high precision assembly devices that must tolerate high temperature environment. Such type,

classified as 900-series superalloy, consisted primarily of Incoloy or GH 903, 907, 909 containing systematic variations in Fe, Nb, Si and C [1]. In addition, the partial or complete substitution of Nb for Cr accounts for the extremely low thermal expansion of such materials at high temperatures [2]. On one hand, there emerged a need to join two or more component

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parts as device structural complexity increased [3]. On the other hand, it was of considerable importance to clarify the weldability and metallurgy behavior in successfully joining the components [4], particularly utilizing advanced welding technology. Solidification microstructure and cracking phenomenon including their relations were merely investigated [5] in terms of arc welding. Current research literatures on arc welding of 900-series superalloy includes the liquation mechanism in the heat affected zone (HAZ). Such kind of liquation was controlled by alloy composition and temperature range. Also, these two factors affected microstructural morphology and crack tendency in the fusion zone (FZ). The conclusion was drawn that partial melting and liquation cracking sensitivity was primarily related to the initial fraction of Nb-rich carbides, as well as the resultant Laves distribution and austenite (γ) grain size [6]. By using a pseudoternary γ -Nb-C surface solidification theory, the sequence of 900-series superalloy solidification was manifest in three reaction processes [7], namely (1) primary $L \rightarrow \gamma$, (2) $L \rightarrow (\gamma + \text{NbC})$ and (3) $L \rightarrow (\gamma + \text{Laves})$. Concretely, reaction (1) revealed that γ was the first to precipitate with rejection of Nb, Ti and C into interdendritic regions. In reaction (2), NbC materialized and simultaneously consumed surrounding C atoms, resulting in the subsequent eutectic formation of Laves in reaction (3). Therefore, the morphology of primarily formed γ grains dominated, or more precisely shaped, the appearance of γ /Laves eutectic constituents to a large degree.

For relatively reliable connectivity, the frontier fusion joining technology was adopted recently, including traditional arc welding and newly developed high-energy beam welding. So far, laser welding technology has proven to be superior to traditional arc and electron beam welding in heat input, welding speed, less damage to base metal (BM) and extensive flexibility [8]. Recently, visible defects, such as pores and cracks, were common to generate in GH909 weldments while unitizing laser manufacturing technology [9]. The microstructure, porosity, and strengthening forms together resulted in the softening behavior of the resultant joint [10]. In the previous study, the inhabitation of HAZ liquation cracking was achieved by essentially reducing laser power input, which determined the heat input [11]. In addition, the crack formation during laser welding GH909 was caused by a three-step process including intergranular liquation, then crack initiation and final crack growth [12]. MC-type carbides (e.g. NbC) and eutectic phases (e.g. γ /Laves eutectic constituents) were accommodated at interdendritic regions due to compositional segregation during weld pool solidification [13]. Also, the microstructural variation from a cellular structure to a dendritic and even minor equiaxed [14]. We conclude from these results that laser beam welding has proven to be a feasible method for welding GH909 superalloy. Notably, weldment quality was highly dependent on microstructure. Although Laves phase or γ /Laves constituents accounted for only a minor fraction, their distribution and morphology affected the mechanical properties [15]. Hence, the weldment microstructure sequentially solidified from fusion line (FL) to weld center played an essential role in the service life of the welded joint. Generally, there exists four trunk types in as-solidified castings or weldments, namely planar, cellular and dendritic morphologies [16]. When applying different solidification conditions, the

resultant shape can evolve from one to another [17]. At the meantime, microstructure was closely related to the mechanical performance, such as microhardness [18]. In summary, a better understanding of the solidification process and key contributing factors during laser welding of 900-series superalloy is now a matter of considerable importance.

In this study, we investigated the solidification features of GH909 alloy. From FL to weld center, the following characteristics were the subject of analysis: 1. The morphological distribution of γ trunks and γ /Laves constituents; 2. Growth mode and corresponding evolution process from FL to weld center; 3. Crystalline features including orientation, Schmidt factor, grain boundary and size of γ grains, as well as orientation relationship between γ and Laves phases; 4. Forming mechanisms of γ trunks and interdendritic Laves.

2. Experimental setup

For this experiment, a 2 mm-thickness GH909 superalloy plate was utilized. The relevant normalized chemical compositions were listed in Table 1. As stated, GH909 superalloy is a Fe–Co–Ni based, Nb-bearing and precipitation-strengthened type superalloy. Laser welding technology was employed due to its proven efficiency and quality. Concretely, a Fiber laser system (type: IPG YLR-4000) with a 0.5 mm-diameter beam spot at the focal plane was the equipment of final choice. A schematic diagram of laser welding and thermal monitoring (type: FLIR) was shown in Fig. 1. To avoid high reflectivity, laser beam was deliberately deflected from vertical direction with 10° . Other welding parameters were displayed in Table 2. While recording the temperature profile during solidification, the threshold was set as 2100°C with a sampling frequency of 30 Hz. Prior to welding, linear cutting and laser cleaning were conducted to ensure that butt end faces were smooth, aligned and uncontaminated.

After completing laser butt welding, a representative sample was cut cross-sectionally from the middle position of the welded seam. The sample was subjected to a typical and integrated coarse-to-fine grinding procedure in order. Then, electrochemical polishing was used to eliminate the potentially uncontrollable effects of near superficial residual stress on follow-up electron microscopy tests. The electrochemical polishing and etching solutions were comprised of $\text{HClO}_4 + \text{C}_2\text{H}_5\text{OH}$ (volume ratio 1:9) and $\text{HCl} + \text{HNO}_3 + \text{H}_2\text{O}$ (volume ratio 3:1:4), respectively.

Having the appropriate preparations listed above, optical and electronic observations were performed on the sample including optical microscope (OM), scanning electron microscope (SEM), energy dispersion spectrum (EDS) and electron back scattering diffraction (EBSD). These metallurgical analysis devices were used to analyze macro/micro structural morphologies, composition profile, crystalline orientations

Table 1 – Chemical composition of GH909 superalloy (wt. %).

Element	Fe	Ni	Co	Nb	Ti	Si
Base metal	40.88	38.25	14.25	4.59	1.66	0.37

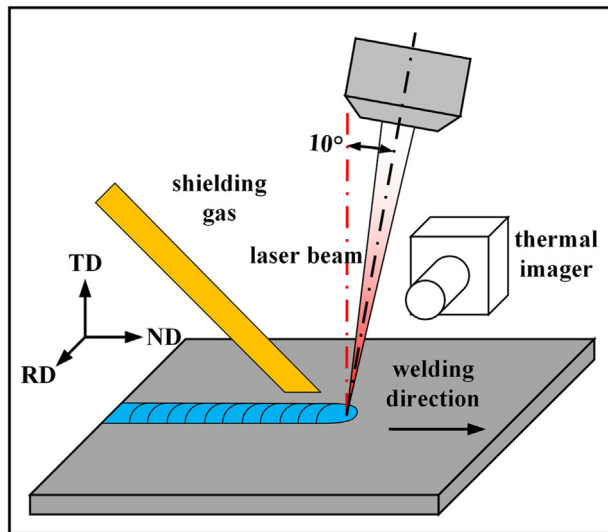


Fig. 1 – Schematic diagram of welding process equipped with monitoring device.

and other derived features using ATEX [19]. During conducting SEM analysis, secondary electron (SE) and back scattering electron (BSE) modes were collected. In common practice, SE is the preferred mode when the surface to be analyzed exhibits abundant structural details generated from grinding or corrosion. Nevertheless, BSE is preferentially adopted when compositional and orientational variations in local regions dominate [20]. Inasmuch as γ /Laves constituents merely accounting for a relatively minor fraction, Extreme-EDS, was utilized in this study to minimize the size of beam-affected zone to more precisely depict the compositional variations. To achieve such purpose, parameters of 5 kV accelerating voltage and a 4 mm working distance were utilized.

Seeing the dominant fraction of γ grains in the previous study [21], Schmidt factor was calculated using ATEX software to evaluate the slip tendency of γ phase, assessing its natural resistance to deformation under forces with specific directions. As stated by previous researchers and Eq. 1.1, when the critical resolved shear stress exists in a polycrystalline material, the presence of relatively lower SF in a local region lowered the yield strength, indicating a “soft” zone favorable for generation of dislocations, cracks and failure. Therefore, it is essential to characterize the SF distribution in the weld joint.

$$m = \tau / \sigma \quad (1.1)$$

where m was the Schmidt factor, σ was the yield strength and τ was the critical resolved shear stress.

3. Microstructure and microtexture

3.1. Evolution from FL to weld center

When evaluating the welding metallurgy for certain materials, the first consideration is the integrity of a welded joint on macro and micro scales, namely absence of fabrication-related defects. In the present study, the laser butt welded joint of a 2 mm GH909 plate in the transverse section was presented with a “Y” type FZ in Fig. 2. As previously stated, the setting a 0 mm defocusing distance on the top surface resulted in a nearly parallel 0.5 mm-diameter beam considering a height variation of 2 mm. In Fig. 2a, the width of the straight-line region was measured as 0.77 mm, slightly larger than the beam diameter of 0.5 mm. In other positions, however, an obvious deflection of FL from the vertical direction was observed at the top and bottom with a longer width in the former. This phenomenon was primarily attributed to the presence of Marangoni force induced convection [22]. Relatively intensive molten pool convection at the top led to an apparently wider width of 1.46 mm than the bottom of 0.90 mm. Fig. 2b exhibited the same cross-sectional morphology but in BSE mode. Obviously, it depicted two distinct features between BM and FZ. As can be seen, the former primarily consisted of fine and equiaxed grains while the latter was characterized by the coarse and columnar. FZ can be further subdivided into two portions, equiaxed and columnar zones. Additionally, there exhibited an apparent morphological inhomogeneity between BM and FZ, particularly near to the FL. For clarification, two enlarged regions covering both deflected and vertical FLs were chosen and analyzed using electron microscopy in Fig. 2c and d.

Fig. 2c featured a region where FL is straight but apparently off vertical. In FZ, epitaxial growth of γ trunks (gray appearance) were initiated from FL and exhibited various growth directions. Such appearance originated from the naturally differed $\langle 100 \rangle$ crystalline orientations in the substrate (BM). As the molten pool started to solidify, the straight FL performed as a solid/liquid interface. In this case, a nearly unidirectional maximum temperature gradient (∇T_{max}) was produced perpendicular to the FL, presenting a parallel but inverse direction to the optimal heat diffusion. Therefore, γ trunks having growth directions close to such direction will be promoted after nucleation. As for γ /Laves constituents (white appearance), they were morphologically trapped in the interdendritic regions of adjacent γ trunks. Immediately adjacent to FL, there exists a fairly narrow region where γ trunks exhibited a planar growth fashion. In this region, there existed no apparent dendrite features except for dispersedly distributed Laves phase. After this, dendritic γ trunks were observed. In BM, or more precisely HAZ, typically inter- and intra-granular liquation behavior was shared. The reasons for the above two liquation phenomena are primarily attributed to three mechanisms. The first one was a secondary phase (e.g. NbC, TiC, etc.) partial dissolution effect (zone 1) [23]. The second was a Nb rich solute “pipeline” diffusion effect (zone 2) [24]. The last was grain boundary wetting effect (zone 3) [12].

In the vertical FL displayed in Fig. 2d, liquation appearance in HAZ kept almost unchanged but microstructural evolution

Table 2 – Laser welding variables.

Equipment	Defocus distance/mm	Laser power/kW	Welding speed/(m·min ⁻¹)	Shielding gas flow/(m ³ ·h ⁻¹)
YLR-4000	0	1.5	1.5	Pure Ar, 1.5

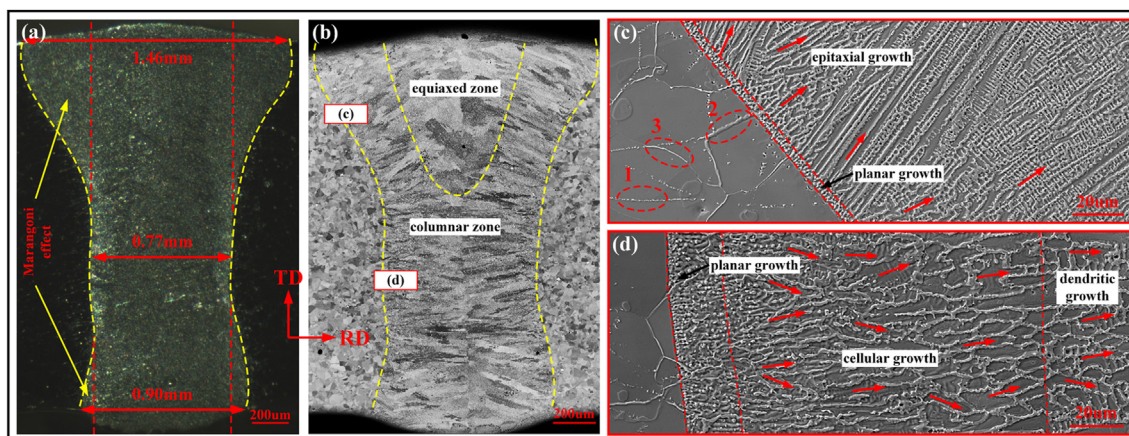


Fig. 2 – An overview of the GH909 laser-welded joint on macro and micro scales. (a) and (b) were cross-sectional macro morphologies obtained by OM and BSE, respectively. (c) and (d) were amplified zones within in (b) to reveal microstructural features. These areas were located near the slant and vertical FL shapes at the top and middle positions. Note that the red arrows in (c) and (d) exhibited the alignment or growth direction ($\langle 100 \rangle$ -crystal direction) of γ trunks.

varied significantly from FL to weld center. Obviously, there emerged a progressive increase in the length of planar growth encountered in Fig. 2c. Even after this stage, the subsequent feature of γ trunks was cell-shape rather than the dendrite. During cellular growth, the primary spacing (λ_1) gradually increased as the distance from FL. Not until the distance from FL increased to a critical value was the dendritic growth encountered. In the whole process of microstructure variation, the horizontally arranged ∇T_{max} was probably the cause for continuous growth of planar, cellular and dendritic γ trunks. The primary evidence lies in the alignment of $\langle 100 \rangle$ -crystal directions along ∇T_{max} .

As observed and discussed in Fig. 2d, γ phase and resultant γ /Laves constituents exhibited distinct morphologies and alignments throughout planar, cellular and dendritic stages. Two BSE pictures shot along horizontal direction but with different heights were chosen to clarify the transition feature between adjacent stages from FL to weld center, as depicted in Fig. 3a and b. As can be seen, zone 1 and zone 3 were near to the planar/cellular borders while zone 2 and zone 4 the cellular/dendritic borders.

In the two BSE pictures, the white regions identified the γ /Laves constituents while the gray γ trunks. The presence of γ /Laves constituents was most frequently encountered at the interdendritic regions of γ trunks regardless of growth modes. Therefore, γ trunks was observed to decide the solidification of γ /Laves constituents by morphology and location limitations. In terms of morphology, γ /Laves constituents as well as γ trunks varied considerably in planar, cellular and dendritic zones, markedly depending on the distance from FL. In the planar zones, γ /Laves constituents were trapped in γ matrix with a dispersed and discontinuous form. As progressing to the cellular zones, γ trunks were changed to cellular morphology on the basis of its lamellar growth originating from central position. Simultaneously, γ /Laves constituents divaricated and evolved into a reticular structure. Then, at the dendritic zones, secondary branches of γ trunks came into being and gradually developed, resulting in dendritic γ trunks

and γ /Laves constituents. Furthermore, it was worth noting that two distinct types of γ trunks were observed in this stage. One was the cellular γ dendrites, the other was the “real” γ dendrites. In the former situation, the tips of both trunks and branches were characterized by lamellar growth and elliptical borders that was observed in cellular growth. During the whole process, no obvious NbC particles was observed due to a low C and high Nb composition feature [7].

In zone 1 and zone 3 (red dotted circles), noodle-shaped γ trunks progressed across the planar/cellular interface with gradually widening primary spacing. Note the fragmentation occurring at the tip site of one γ trunk in zone 1 and then it formed a typical cell. Such detachment behavior produced newly oriented γ and blocked the continuous growth of the previous trunk. Another phenomenon needed to notice was newly nucleated γ trunks with various growth directions in zone 3. Despite a consistent cellular morphology, internal composition inhomogeneity in these γ trunks was presented in dendritic fashion. This indicated the trend to form dendrites at cellular stage.

In zone 2 and zone 4 (yellow dotted circle), a cluster of γ trunks was encountered with lamellar and elliptical composition inhomogeneity at cellular stage. After evolving into dendritic stage, such structure was transformed into more branched and more widely-spaced dendrites. Considering three isolated γ trunks ranging from the small and equiaxed to the fully developed, fragmentations also occurred and eventually they grew into dendrites. Note that composition displayed almost a more uniform distribution in γ dendrites than γ cells. In addition, γ primary spacing exhibited a slight increase from 1.92 μm in planar stage to 3.64 μm in the cellular but a progressive increase from the cellular to the dendritic (12.62 μm). These means dendritic zone was dominated by coarsely-sized, well-branched and widely-spaced trunks.

Fig. 4 depicted a linear extreme-EDS scanning outcome on a micro scale. It was intentionally conducted to illustrate the specific redistribution behaviors of major elements among γ branches in terms of Fe, Ni, Co, Nb and Ti. Like before, the

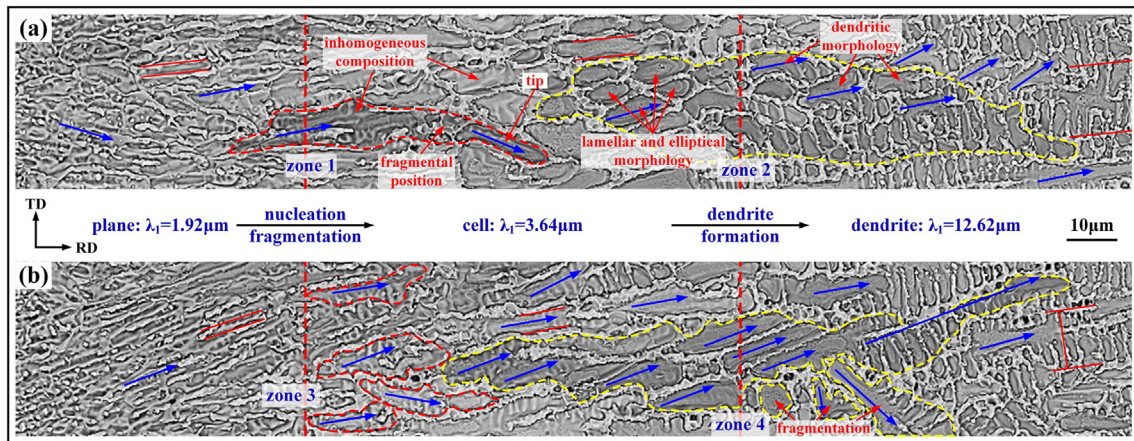


Fig. 3 – Morphological evolutions from planar to cellular and dendritic stages along RD axis but with different height along TD axis.

white region was supposed to represent Laves phase while the black γ trunks. Overall, the content of Fe displayed a sinusoidal variation with tips locating at the γ trunks and depressions the Laves phases. Like the content of Fe, both Nb and Ti exhibited similar distribution but with completely inverse arrangements of tips and depressions. In addition, compared to Nb, Ti possessed a progressively low amplitude of variation. The distributions of Ni and Co kept almost unchanged throughout the γ and Laves structures. Such kind of distribution conformed to the previous published results regarding welding superalloy 903 [25]. Based on the above results, equilibrium was assumed to exist between primary Fe and complementary Nb and Ti to some degree. In other words, an increase in Fe content was inevitably accompanied by an approximately equal decrease in the sum of Nb and Ti, as listed in Table 3. As can be seen, Mn and Si, although acting as minor elements, segregated to Laves phase. The composition profile displayed above further confirmed that Laves did not

form until the Nb content rose over a critical level in the middle of two γ trunks. Moreover, qualitative analysis of the mass content in γ trunks and Laves phases figured out a progressive variation in partition coefficient (k) of Fe, Nb and Ti. Here, k was calculated based on the ratio of mass content in γ trunks to that in Laves phase considering the precipitation priority of the former phase. In this means, $k_{Fe} = 1.56$, $k_{Nb} = 0.12$ and $k_{Ti} = 0.34$. Lattice parameters and schematic configuration of γ and Laves phases were also listed in the forms of face centered cube (FCC) and close-packed hexagon

Table 3 – Chemical compositions of γ and Laves structures (wt. %).

Structure	Fe	Ni	Co	Nb	Ti	Mn	Si
γ	43.69	37.58	14.83	2.05	1.10	0.50	0.25
Laves	28.09	37.38	12.90	16.94	3.22	0.60	0.87

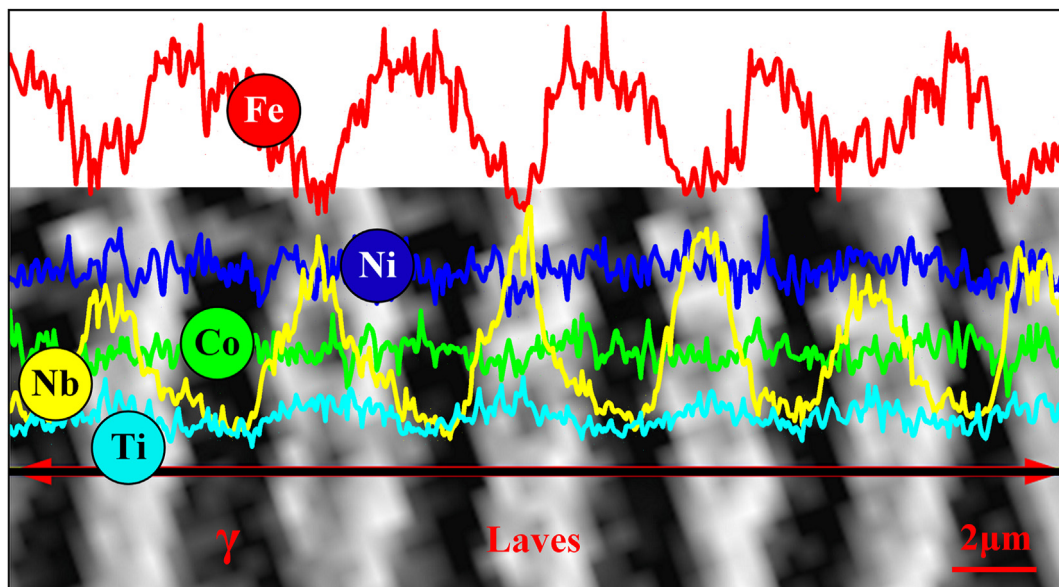


Fig. 4 – Compositional profile characterized by Extreme-EDS through a horizontal line.

(HEX), respectively, as shown in Table 4. Note that the latter was representatively depicted by a third of the whole structure considering its geometric rotation symmetry.

Based on the crystalline information listed above, EBSD was conducted at a relatively tiny region to facilitate detection of γ and Laves phases. Band contrast (BC) map depicting an entire morphology covering unfused BM and half of FZ. The gray value of the BC map represented the signal strength of the Kikuchi line, indicating the deviation of practical lattice arrangement from the ideal one in database. The appearance of gray and bright regions demonstrated the progressively different Kikuchi line signals between γ and γ /Laves eutectic constituents, as indicated from Fig. 5a. Concretely, the former possessed relatively higher gray value than the latter in BC map. As Kikuchi line signal collected by EBSD probe was immediately relevant to the order degree of lattice arrangement. Therefore, γ /Laves constituents exhibited an essentially disordered and distorted atom configuration. Furthermore, considering the gray value profile of the BC picture, γ and Laves can be differentiated in their eutectic constituents. To distinguish the γ in eutectic constituents from γ trunks, the former was called as “eutectic γ ”. Hence, the scopes of gray values for γ , eutectic γ and Laves structures primarily arranged from 25 to 90, from 110 to 150 and from 170 to 190, respectively. Corresponding proportions turned out to be 77.96%, 17.22% and 4.82%, respectively. Therefore, gray values lower than 150 can be recognized as γ /Laves constituents in a BC map. As can be inferred from Fig. 4, eutectic γ immediately

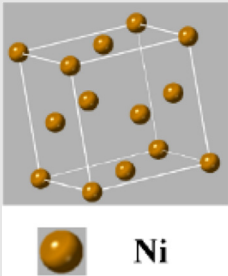
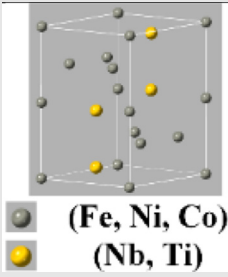
adjacent to Laves contained a higher Nb content than γ trunks but lower than Laves. We concluded that the presence of Nb atoms in γ phase probably resulted in an inconsistent and curved lattice arrangement. Like BC map, GND density was another feature reflecting the lattice arrangement. As can be seen, eutectic γ near Laves displayed a relatively intensive GND density compared to γ , as shown in Fig. 5b. Notably, the fraction of high GND density ($>7.5 \times 10^{14} \text{m}^{-2}$) agreed well with that of the gray value from 110 to 150, verifying the stated effect of Nb in eutectic γ . The relatively higher GND density was also located at and near the sharp edge due to the curving phase boundaries [26].

Generally, due to the eutectic solidification phenomenon, some crystalline orientation relationships were likely to be established at phase interfaces [27]. As shown in Fig. 5c, $\{1\ 1\ 0\}$ and $\{1\ 1\ 1\}$ pole figures of γ phase were plotted. Similar treatments were also employed in Laves phase for $\{1\ 1\ 2\ 0\}$ and $\{0\ 0\ 0\ 1\}$ planes. Comparing the overlap positions of these four pole figures, we confirmed that Shoji-Nishiyama (S.-N.) orientation relation [28], namely $(1\ 1\ 1)_{\text{fcc}} // (0\ 0\ 0\ 1)_{\text{hex}}$ and $[1\ 0\ -1]_{\text{fcc}} // [1\ 1\ -2\ 0]_{\text{hex}}$, were established at the phase boundary of Laves and γ , or more precisely, eutectic γ . The above outcomes in return the eutectic features of γ and Laves at the interdendritic regions.

As described above, Marangoni force and deflected FL exerted complicated and unquantifiable influence on epitaxial growth of γ trunks. Therefore, for convenience, a micro region near the straight and vertical FL (Fig. 2d) was chosen for further EBSD test within a large zone, covering BM (actually HAZ but called as BM in later statements) and epitaxial growth from FL to weld center, as shown in Fig. 6. The solidification process from FL to weld center mainly experienced initially planar, then cellular and finally dendritic stages. The length of these three zones were measured as 62 μm , 125 μm and 200 μm , respectively. Note that in Fig. 2a, the entire width (774 μm) is approximately equal to 0.77 mm (Fig. 2a), indication the reliability of EBSD test. Note that γ phase was set as the only one that can be recognized during EBSD testing. In this way, eutectic γ and Laves structures was assumed to exhibited relatively lower gray value in the resultant BC map. Particularly, in BM, grain boundary exhibited a low gray value feature as it was naturally comprised of disordered atom arrangement. In FZ, γ /Laves constituents existed at both interdendritic regions and grain boundaries [21]. Based on the differences in gray value between γ and γ /Laves constituents, the fraction of γ trunks can be estimated by calculating the gray value over 150 in planar, cellular and dendritic zones. It should be noted that BM possessed no γ /Laves constituents, suggesting the relatively high effect of grain boundary (1.88%). From planar to dendritic zone, the fraction of γ first decreased ($99.22\% \rightarrow 97.37\%$) and then slightly increased ($99.22\% \rightarrow 97.37\%$). Inversely, the fraction of γ /Laves constituents experienced an increase from planar to cellular zone and then a decrease from cellular to dendritic zone. During the whole process, planar zone was equipped with the least volume of γ /Laves constituents. Such occurrence corresponded to the morphology of γ /Laves constituents in Fig. 3.

The above observations demonstrated that γ trunks manifested an organized variation from BM to dendritic zone.

Table 4 – Crystalline parameters of γ trunks and Laves structures.

structure	Lattice parameters	Lattice angle	Configuration
γ	a = 0.357 nm b = 0.357 nm c = 0.357 nm	$\alpha = 90^\circ$ $\beta = 90^\circ$ $\gamma = 90^\circ$	
Laves	a = 0.523 nm b = 0.523 nm c = 0.857 nm	$\alpha = 90^\circ$ $\beta = 90^\circ$ $\gamma = 120^\circ$	

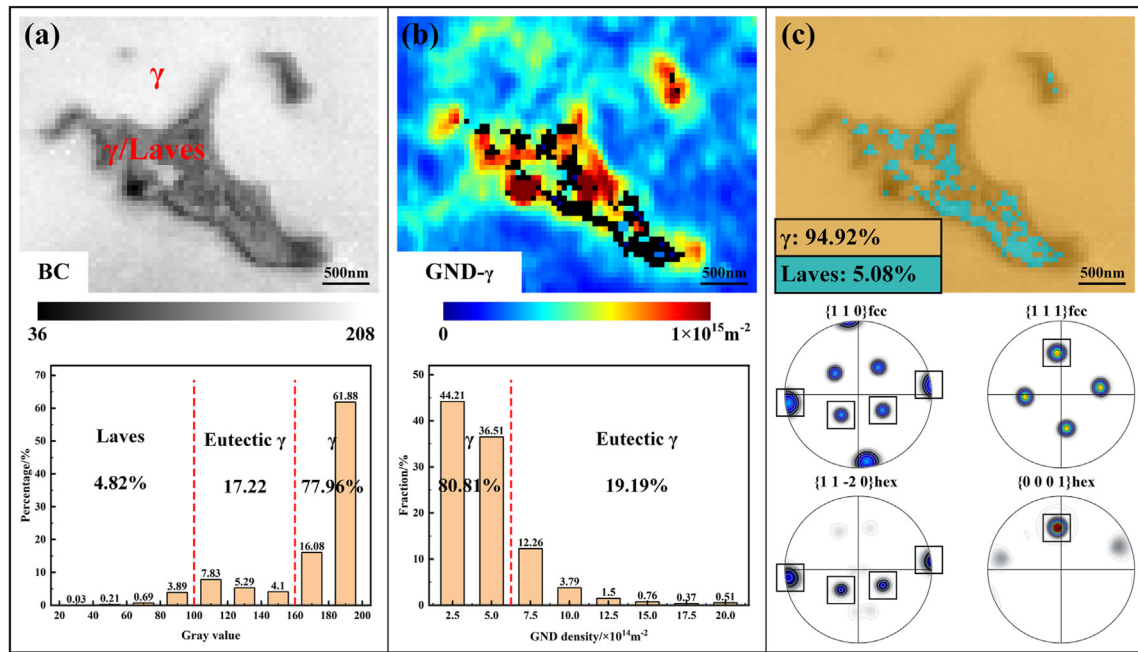


Fig. 5 – Crystalline structure existing in γ and γ /Laves constituents. (a) BC map and corresponding profile; (b) GND density map and corresponding profile; (c) phase map and pole figures estimating the possible FCC-HEX orientation relationship.

Further clarification of their growth features during the solidification process was also of considerable importance. An integrated IPF figure derived from RD axis helped to analyze crystalline orientation and texture variations between BM and FZ, as well as evolution from planar to dendritic zone, as displayed in Fig. 7a. $\{100\}$ -Pole figures at RD-TD plane (called as $\{100\}$ -PF), inverse pole figures along RD axis (called as IPF-RD) and orientation distribution figure (called as ODF) were plotted in Fig. 7b-d. As decided by characterization methods, PF was advantageous in characterizing the sheet texture while

IPF the fiber type. In cubic crystal, the $\langle 100 \rangle$ -crystal directions, also called as growth direction, were naturally perpendicular to $\{100\}$ -crystal planes. Therefore, PF was also able to reflect the distribution of $[100]$ crystal direction of γ trunks in sample coordinate system. ODF was chosen due to its unique advantage while demonstrating a three-dimensional (ϕ_1 , Φ , ϕ_2) orientation distribution. In this study, texture was marked by Miller index near the maximum pole density points at the specific section at $\phi_2 = 45^\circ$ with ϕ_1 and Φ ranging from 0° to 90° .

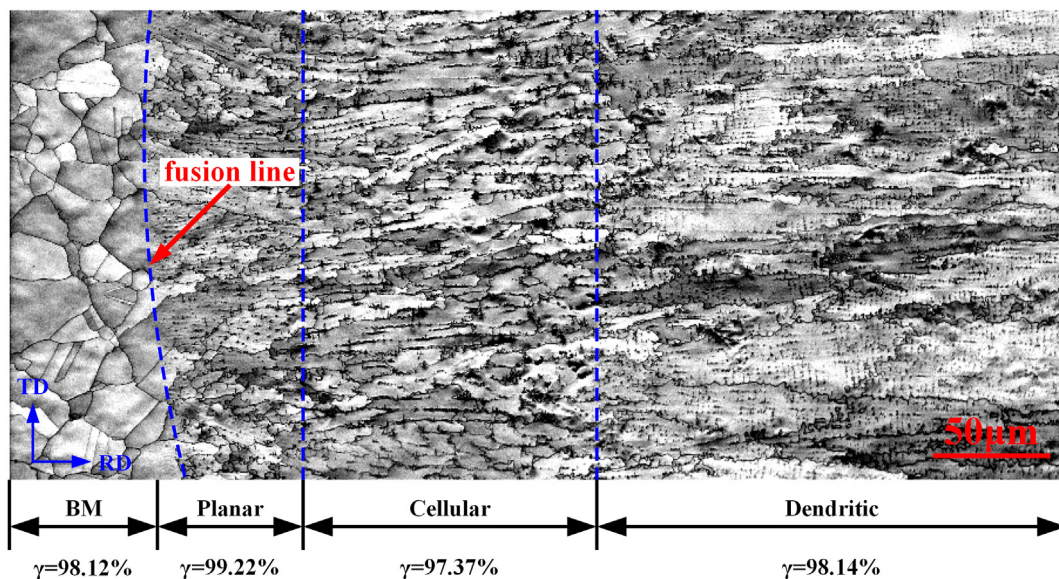


Fig. 6 – Morphology and fraction of γ trunks from BM to weld center in Fig. 2d. (a) was an overall BC map with manually differentiated subregions, BM and FZ. According to the morphological features, FZ was divided into planar, cellular and dendritic zones.

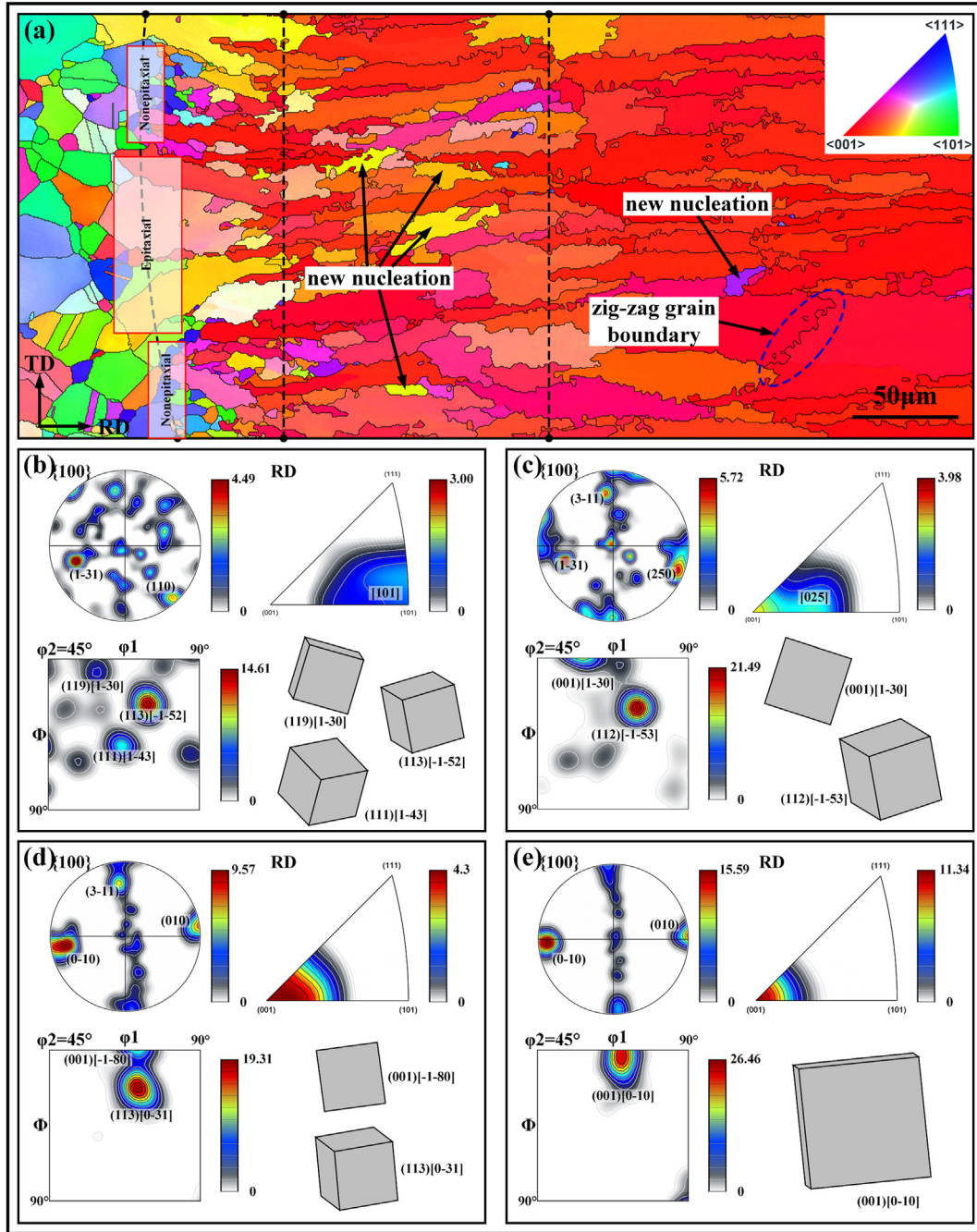


Fig. 7 – Transformation of crystalline texture in γ grains from BM to dendritic zone. (a) IPF figure projected along the RD direction; (b)–(e) detailed texture corresponding to $\{1\ 0\ 0\}$ -PFs, IPFs-RD and ODF figures at $\phi_2 = 45^\circ$ including representative cubes oriented in subsets BM, planar, cellular and dendritic.

First consider BM, essentially functioning as a substrate for epitaxial growth. Overall, BM exhibited a usual appearance of randomly oriented and nearly equiaxed γ grains after rolling, solution and annealing procedures. Therefore, $\{1\ 0\ 0\}$ -PF displayed a dispersed distribution with relatively high density at $[1\ 1\ 0]_{\text{sample}}$ and $[1\ \bar{7}\ 9\ \bar{3}1]_{\text{sample}}$. IPF-RD also possessed a concentration at $[1\ 0\ 1]_{\text{sample}}$. Hence, ODF shared a dispersed arrangement of $(1\ 1\ 9)$ $[1\ \bar{7}\ 9\ \bar{3}0]$ $(1\ 1\ 3)$ $[\bar{1}\ 5\ 2]$ and $(1\ 1\ 1)$ $[1\ \bar{4}\ 3]$.

Then the planar stage, it was partially characterized by epitaxial growth and residually by the non-epitaxial. Therefore, epitaxial behaviors of γ grains were observed with various lengths, primarily dominated by competition growth among favored and unfavored growth. During this stage $[1\ 0\ 0]$ -crystal directions mainly centered at $[3\ \bar{7}\ 9\ \bar{1}1]_{\text{sample}}$ $[1\ \bar{7}\ 9\ \bar{3}0]_{\text{sample}}$ and $[2\ 5\ 0]_{\text{sample}}$ in $\{1\ 0\ 0\}$ -PF. These directions were either succeeded from parent grains in BM or generated from newly nucleated grains, both gradually getting parallel to

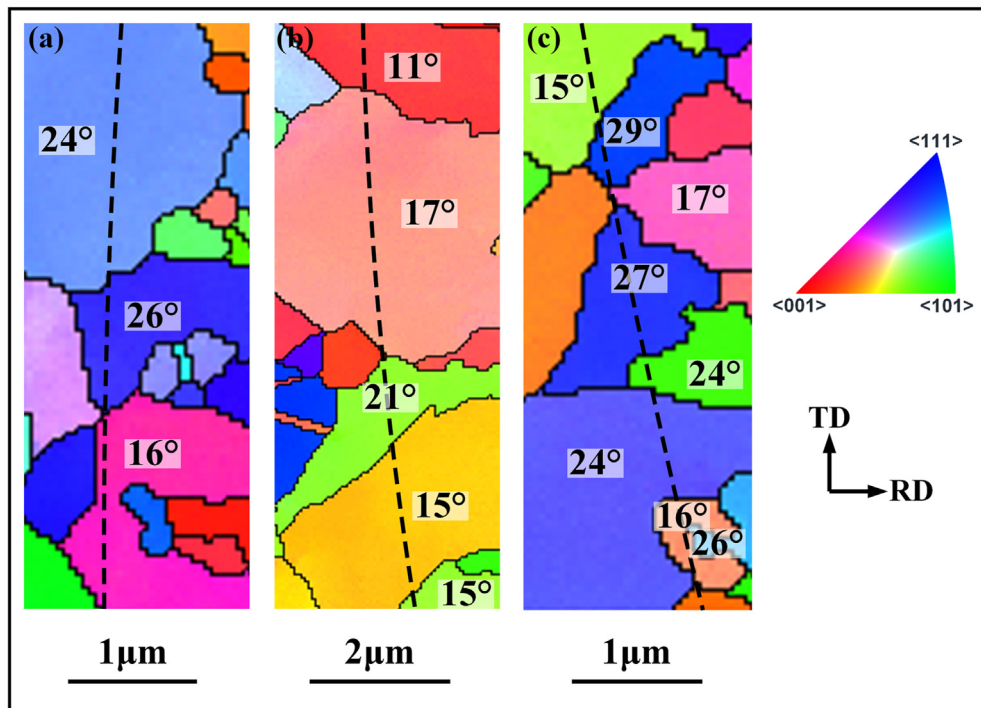


Fig. 8 – Spatial angles between $[1\ 0\ 0]$ -crystal direction and RD axis of γ grains along FL concerning epitaxial growth (b) and non-epitaxial growth (a, c).

RD axis. Resultantly $[0\ 2\ 5]_{\text{sample}} + [0\ 0\ 1]_{\text{sample}}$ in IPF-RD and $(0\ 0\ 1)\ [1\text{--}7,\ 9\text{--}31] + (1\ 1\ 2)[-1\ -5\ 3]$ in ODF presented a similar feature.

Subsequently, in cellular stage $[1\ 0\ 0]$ -crystal directions were increasingly parallel to RD axis in terms of $[0\ 1\ 0]_{\text{sample}}$ and $[0\text{--}1\ 0]_{\text{sample}}$ in $\{1\ 0\ 0\}$ -PF. It should be noted that $[0\ 1\ 0]$ - and $[0\ 0\ 1]$ -crystal directions, acting as side branches, were dispersedly but homogeneously located along TD axis. Such behavior demonstrated a rotation of the latter two crystal directions along the axes of primary trunks. Concentration at $[0\ 0\ 1]$ in IPF-RD and $(1\ 1\ 3)[0\text{--}3\ 1] + (0\ 0\ 1)[-1\ -8\ 0]$ in ODF further exhibited the same growth characteristic.

Finally, consider the dendritic stage. From $\{1\ 0\ 0\}$ -PF and IPF-RD, this region exhibited a strictly unidirectional growth fashion with $[1\ 0\ 0]$ -crystal directions aligning along RD axis. Meantime $[0\ 1\ 0]$ - and $[0\ 0\ 1]$ -crystal directions were distributed in a progressive dispersed feature along the axes of primary trunks. Besides, texture presented an apparent assembly at $[0\ 0\ 1]$ in IPF-RD $(0\ 0\ 1)[0\text{--}1\ 0]$ in ODF. Noticeably, the maximum pole density of PF, IPF and ODF were continuously increased from BM to dendritic zone. Morphologically, zig-zag grain boundaries were only observed in dendritic zone. Such phenomena were primarily attributed to the touching and interacting behaviors of freely nucleated grains with similar growth directions but different orientations. In detail, the side branches intrude into the micro gaps existing between the secondary dendritic arms [29].

Based on the existence or length of epitaxial growth from BM, the planar stage can be divided into non-epitaxial (Fig. 8a and c) and epitaxial (Fig. 8b) zones. It was concluded that γ grains whose $[1\ 0\ 0]$ -crystal directions were aligned along RD axis possessed a relatively longer epitaxial crystal growth

than those not. Under the latter situation, growth inhibition and alteration were frequently encountered. Note that RD axis was basically parallel to the heat diffusion direction. Therefore, the minimum value of spatial disorientations between three $\langle 1\ 0\ 0 \rangle$ -crystal directions and RD axis was defined as the angle between $[1\ 0\ 0]$ -crystal direction and preferred growth direction. Overall, non-epitaxial zone was comprised of relatively higher angles ($\sim 24^\circ$) than the epitaxial ($\sim 16^\circ$) type. As investigated in directional solidification technology, the inclination of $[1\ 0\ 0]$ -crystal direction to RD (also called heat diffusion direction) promoted the continuous growth for a certain length.

Having global and local discussions, several crystalline orientation determining features were plotted, including Schmidt factor (SF), grain boundary (GB) and grain size (GS), etc. SF was obtained by considering the relative positions between slip plane–direction pair and the force loading direction. Generally, crystals with maximum SF value belong to the easy-to-slip type due to inadequate resistance to deformation, producing considerable localized dislocations and stress concentration. Such regions later acted as an initial fracture point when dislocation density reached a critical level. Generally, SF can be calculated via Eq. 3.1 [30]:

$$m = \cos\phi \cdot \cos\lambda \quad (3.1)$$

where ϕ and λ are the angles between direction of the load force and normal of slip plane $\{1\ 1\ 1\}$, and direction of slip $\langle 1\ 1\ 0 \rangle$, respectively. Force loading along the RD axis was chosen here due to the tensile test to be performed.

Fig. 9a was an integrated SF map of the entire weld joint. Overall, planar and cellular zones displayed a relatively higher area fraction of high SF than BM and dendritic zone. Fig. 9b

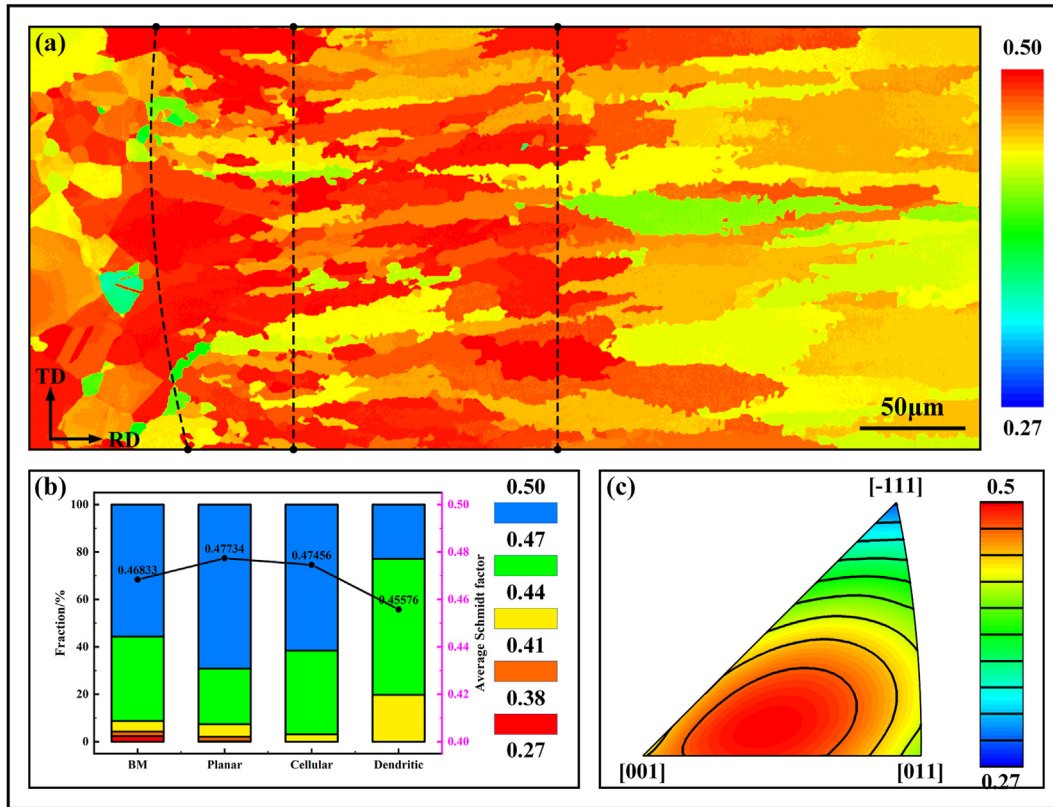


Fig. 9 – Distribution feature and average value of SF in BM, planar, cellular and dendritic zones. (a) An entire SF map based on the force loaded along RD axis; (b) statistical distribution and average SF value variation of four individual region; (c) schematic diagram of the relationship between crystalline orientation in IPF-RD and corresponding SF value.

graphically illustrated detailed area fraction distribution of different SF values in these four subregions. The fraction of SF value ranging from 0.44 to 0.5 in BM, planar, cellular and dendritic zones accounted for 92%, 92%, 97% and 90%, respectively, while the average value was 0.468, 0.477, 0.475 and 0.456, respectively. Therefore, resultant SF of the planar, cellular, BM and finally dendritic zones was listed in a descending order.

Considering the symmetry of cubic lattice, SF can be calculated by confirming the vector product of the pole coordinate (force direction) in IPF-RD with $[1\ 1\ 1]$ (the normal of slip plane) and $[-1\ 0\ 1]$ (the slip direction) crystal directions. A SF label dependent on crystalline orientation in IPF-RD was plotted in Fig. 9c. In this study, SF can be deduced according to the texture point located in IPF-RD. The SF value of three vertices can be calculated as follow. $SF_{\langle -1\ 1\ 1 \rangle} = 0.27$ and $SF_{\langle 0\ 1\ 1 \rangle} = SF_{\langle 0\ 0\ 1 \rangle} = 0.41$. A maximum SF value of 0.5 existed at $\langle -1\ 2 + \sqrt{6}\ 5 + 2\sqrt{6} \rangle$ where ϕ and λ simultaneously attained 45° . Note that this point was considerably adjacent to the $[0\ 0\ 1]$ vertex, providing an explanation of why $[1\ 0\ 0]$ -crystal orientation control is so important in single-crystal productions [31]. As can be seen, when a small deflection was established between $[1\ 0\ 0]$ -crystal orientation and RD axis, epitaxial growth may be promoted but with unfavorably high SF, as observed in planar and cellular zones. Combined with the variations of IPFs from BM to dendritic zone in Fig. 7b-e, alteration of

crystal direction parallel to RD axis from $[1\ 0\ 1]$ to $[0\ 0\ 1]$ was doomed to go through a first increase and then a decrease in SF when passing point of $\langle -1\ 2 + \sqrt{6}\ 5 + 2\sqrt{6} \rangle$.

Subsequently, grain boundary was typically classified into low angle grain boundary (LAGB: 2° – 15°) and high angle grain boundary (HAGB: 15° – 65°). In order to distinguish subtle variations in the present study, LAGB and HAGB were further subdivided into LAGB-1 (2° – 5°), LAGB-2 (5° – 15°), HAGB-1 (15° – 55°) and HAGB-2 (55° – 65°). Fig. 10a depicted an entire distribution of grain boundary from BM to dendritic zone. Due to rolling and annealing procedures, BM was supposed to contain a considerable HAGB-2, namely $\Sigma 3$ type twin grain boundary [32]. Additional HAGB-1 and LAGB-2 also accounted for a significant fraction while LAGB-1 was minimally present. The refusion effect resulted in an apparent decrease in $\Sigma 3$, though several remainders retained and even extended to a certain length during epitaxial growth process. Overall, HAGB-1 and LAGB-2 dominated in the FZ having no concern with location variation. Particularly, planar zone was partially characterized by bulk HAGB-1 in epitaxial region and residually by smaller reticular HAGB-1 in non-epitaxial region. Grain misorientation less than 5° usually reflected the residual stress [33]. An increase in LAGB-1 from BM to dendritic zone also suggested the stress was relatively concentrated in the dendritic zone, also forming the last-to-solidify region.

In order to quantitatively describe the evolution features of grain boundary from BM to dendritic zone, four from-point-to-

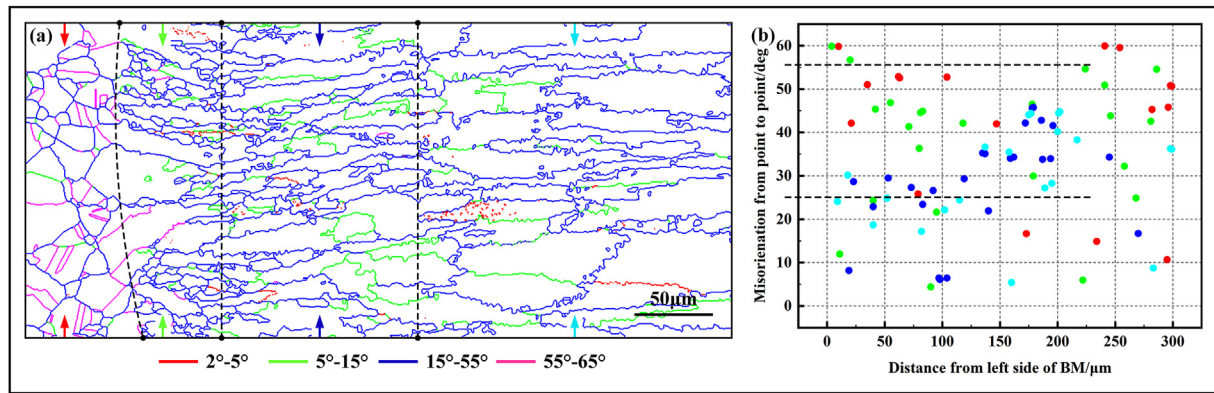


Fig. 10 – Grain boundary map and misorientation profile of grain boundary from BM to weld center. (a) An overall grain boundary map featuring the evolution process; (b) four selected distributions of misorientation from point to point.

point deviation profiles were plotted based on linear scanning at the middle positions (indicated by colored arrows), as plotted in Fig. 10b. The majority of grain boundaries fell between 20° and 50°.

The last feature to be considered in was grain size (GS). Generally, a cluster of γ trunks essentially shared the same crystalline orientation, thus presenting a “single grain” merely with low angle misorientation (less than 15°). Distribution of grain diameter was depicted in Fig. 11a. Overall, BM and planar zone possessed relatively more fraction of small grains than cellular and dendritic zones. The main reason for such occurrence lay in that newly nucleated and randomly oriented grains of tiny size were observed in non-epitaxial region. In Fig. 11b, three size-related parameters (number, maximum size and mean size) were chosen here to investigate the variations among planar, cellular and dendritic zones. Results indicated that the number of grains reached the peak at planar zone while the least at the dendritic zone. The maximum size took on a continuously increased trend from BM to dendritic zone. As for mean size, it present a first decrease from BM to planar zone and then an increase from planar to dendritic zone.

3.2. Forming mechanism

In order to determine the origin of as-solidified microstructure and microtexture from FL to weld center in the above study,

further investigation was carried out on the temperature profile during welding, as shown in Fig. 12a and b. In this study, the thickness of the experimental plate was 2 mm. Hence, the heat diffusion mode can be regarded as a planar type, namely along RD and ND. Based on the microstructure evolution previously studied along RD direction, temperature gradient (G) map was plotted along this direction. The borders of liquid/mushy (L/M) and mushy/solid (M/S) were marked by isotherms of 1360 °C (see dashed line) and 1184 °C (see dotted line) [21], respectively. Besides, full penetration laser welding was expected to induce morphologically similar weld pool along the thickness direction [34]. Therefore, in Fig. 12c, the starting point of planar, cellular and dendritic zones were marked as “A”, “B” and “C”, respectively, in terms of their length ratio observed in Fig. 6, 8, 9 and 10. Obviously, G at planar, cellular and dendritic zones were ranked as: $G_A > G_B > G_C$.

The solidification rate (R) along L/M boundary can be calculated by Eq. 3.2:

$$R = V \cos \theta \quad 3.2$$

where V is the welding velocity, R is the propulsion velocity of L/M, θ is the angle between the welding direction and the normal of L/M. In this way, $\theta_A > \theta_B > \theta_C$ resulted in $R_A > R_B > R_C$, exhibiting a decreasing trend from planar zone to dendritic zone in Fig. 12c. The variation of G, R and corresponding operation of them from FL to weld center was listed in Table 5.

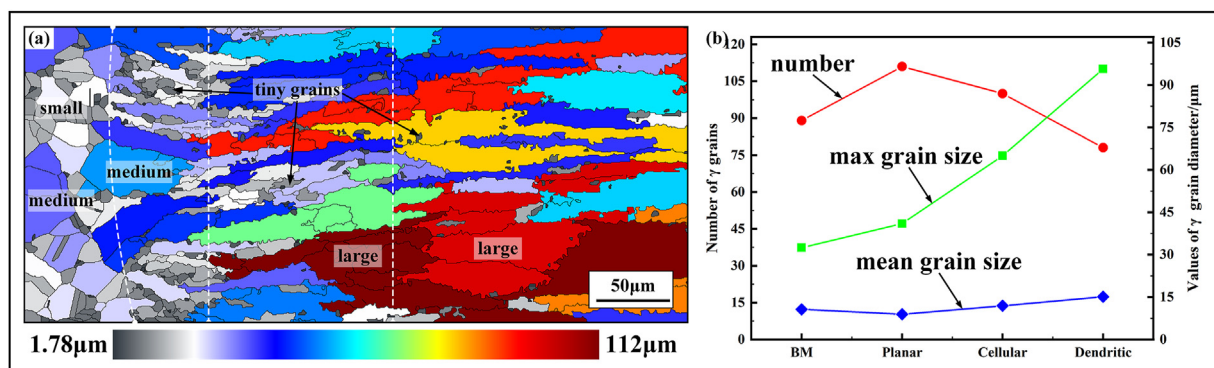


Fig. 11 – Statistical distribution of grain diameter from BM to dendritic zone. (a) Overall distribution; (b) the number of grains and detecting area in BM, planar, cellular and dendritic zones, respectively.

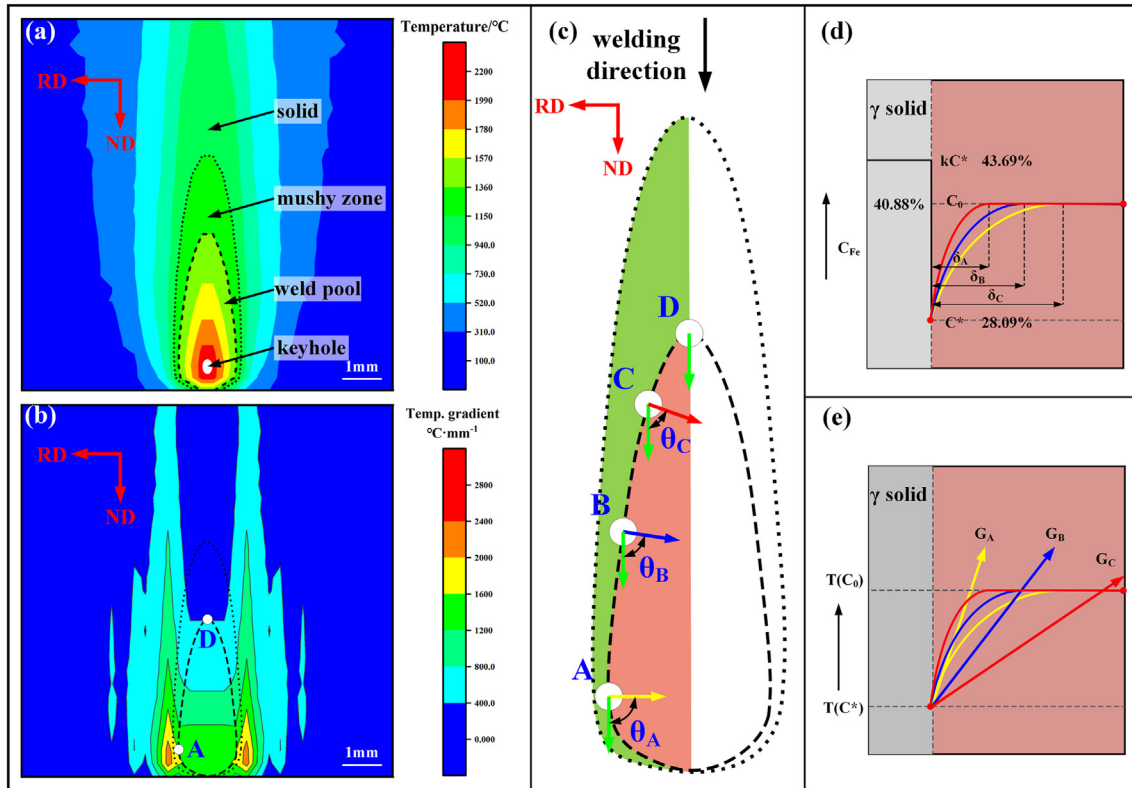


Fig. 12 – Solidification conditions from FL to weld center of the entire joint. (a) was a temperature profile determining the several sub regions; (b) was an absolute temperature gradient along RD axis; (c) was an enlarged view for illustrating variations of solidification rate along FL; (d) and (e) were schematic diagrams for solute distributions of Fe and resultantly distinct supercooling degrees in advance of solidification fronts at planar, cellular and dendritic zones.

During the subsequent growth, there existed several additional factors affecting the formation of solidification microstructure including partition coefficient (k), G and R [36]. As the initial and primary precipitated phase, γ trunks exhibited three differing solidification modes from FL to weld center, namely plane, cell and dendrite. As can be obtained from the previous literatures, the GH909 laser welds were generally accomplished by a two-step solidification process. One was the primary reaction ($L \rightarrow \gamma$) featuring the planar,

cellular and dendritic growth of γ trunks at the L/M boundary. The other was the interdendritic reaction ($L \rightarrow \gamma + \text{Laves}$) characterizing the formation of low-melting-point constituents between L/M and M/S boundaries. In the present study, the former was treated as a single-phase primary growth while the latter poly-phase eutectic growth.

First, consider the single-phase primary growth. According to Fig. 4 and Table 3, the formation of γ trunks was promoted by Fe but inhibited by Nb and Ti in a complementary balance.

Table 5 – Variation of G and R at different stages of weld solidification.

Point	Zone	G ($K \cdot m^{-1}$)	R ($m \cdot s^{-1}$)	G/R	$G^{-1/2}R^{-1/4}$	$G \cdot R$ ($K \cdot s^{-1}$)
A	Planar	1.56×10^6	~ 0	$\sim \infty$	~ 0	~ 0
B	Cellular	8.62×10^5	3.75×10^{-3}	2.30×10^8	4.35×10^{-3}	3.23×10^3
C	Dendritic	5.35×10^5	8.50×10^{-3}	6.29×10^7	4.50×10^{-3}	4.55×10^3
D	Weld center	2.56×10^5	2.50×10^{-2}	1.02×10^7	4.94×10^{-3}	6.40×10^3

Note that the growth direction of γ trunks near the L/M may be not strictly parallel to its propulsion. Therefore, dendrites were easy to degenerate when $[100]$ -crystal directions (preferred growth direction) deflected from ∇T_{max} with an increasing deviation angle [35], leading to the formation of newly oriented tiny grains in Fig. 11a and c. In addition, high G/R largely promoted the epitaxial growth of γ grains whose $[100]$ -crystal directions were inclined to the direction of ∇T_{max} while blocked this behavior when not inclined. Such phenomenon was frequently observed in planar and cellular zones. In the dendritic zone, however, low G/R was unable to block the growth continuity along previously formed $[100]$ -crystal directions along RD axis, as shown in Table 5. Consequently, tiny grains were usually observed in the planar and cellular zones while big grains in the dendritic zone.

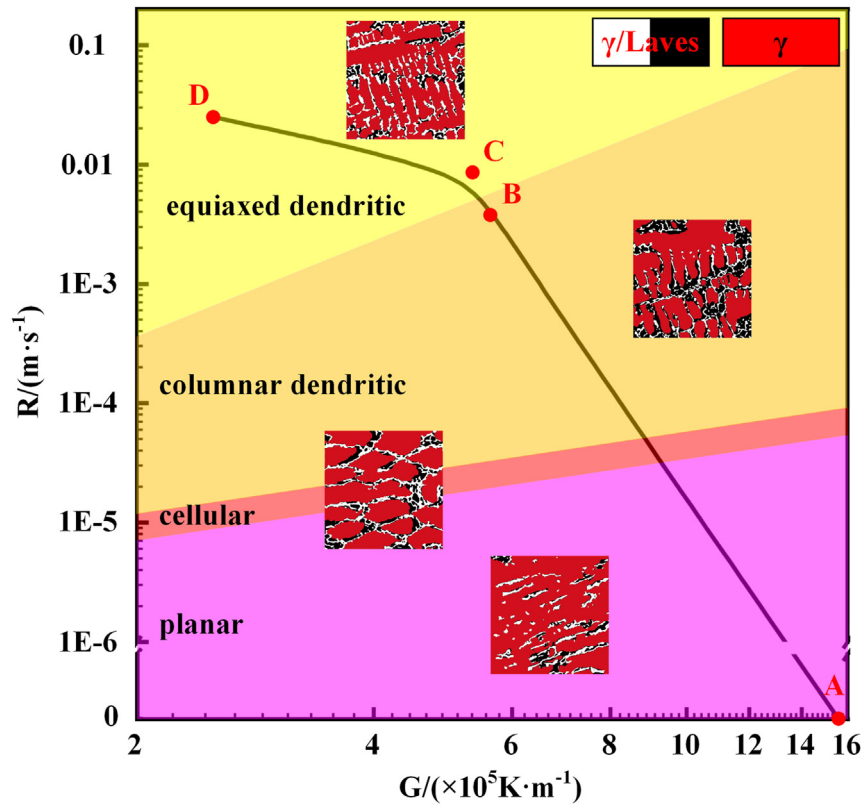


Fig. 13 – Representation of G and R couple and relevant as-solidified microstructures in a schematic summary reprinted from Ref. [39].

For simplicity, the variation of Fe was considered here and k_{Fe} was greater than unity during the solidification of GH909 superalloy. Therefore, Fe solute will dissipate ahead of the L/M boundary (C^* , 28.09 wt. %) due to the higher solubility of the solid ($k_{Fe}C^*$, 43.69 wt. %), resultantly leading to a deficient solute boundary layer. To keep the continuity of solute distribution, Fe solute gradually increased exponentially to remote liquid composition (C_0 , 40.88 wt. %) far ahead of the L/M boundary, as depicted in Fig. 12d. The thickness of “equivalent boundary layer” (δ) can be approximately calculated from Eq. 3.3 [37]:

$$\delta = \frac{2D}{R} \quad 3.3$$

where D is the solute diffusion coefficient of Fe solute.

As the propulsion velocity of L/M boundary increased from planar zone to the dendritic, resultantly producing an increasing order from δ_A to δ_B and δ_C . Generally, the compositional variation ahead of the L/M boundary gave rise to alteration of practical solidification temperature locally calculated from the linear solidus and liquidus, as displayed in Fig. 12e. Assuming the constant melting point (T_C^*) is 1360 °C, the combined consideration of $G_A > G_B > G_C$ and $\delta_A < \delta_B < \delta_C$ gave rise to an increasing degree of supercooling from planar to cellular and dendritic zones. The higher degree of supercooling, the more driving force for preferred crystal growth and nucleation. Moreover, the morphological instability can be figured out according to Eq. 3.4:

$$\lambda_c = 2\pi \left(\frac{D\Gamma}{R\Delta T} \right)^{\frac{1}{2}} \quad 3.4$$

where λ_c is critical unstable wavelength, Γ is the Gibbs–Thomson coefficient, R is the solidification rate and ΔT is the supercooling.

When the wavelength of randomly generated perturbation is greater than λ_c , morphological instability is able to occur and persist in the further growth. Therefore, as R and ΔT were increased, a decrease in λ_c promoted and reserved the amplified perturbations, further accelerating the formation of branches. Hence, branching induced cellular and dendritic structures gradually dominated the growth upon solidification from FL to weld center (Fig. 3). In such case, morphological instability and at the tips of γ trunks will occur. In other words, that the division of G and R determined the morphology of γ trunks due to supercooling [38]. Therefore, with a gradually decreasing G_L , solidification modes evolved from planar to cellular, and finally dendritic.

As for primary spacing (λ_1), it can be figured out by Eq. 3.5:

$$\lambda_1 = A_1 G^{-\frac{1}{3}} R^{-\frac{1}{3}} \quad 3.5$$

where the first term, A_1 , is regarded as a constant. Therefore, λ_1 is actually controlled by the second and third terms, G and R , that can be obtained from Table 5. As calculated, primary spacing, as well as γ grain size, was supposed to display an increasing trend from FL to weld center corresponding to Figs. 3, 6 and 11.

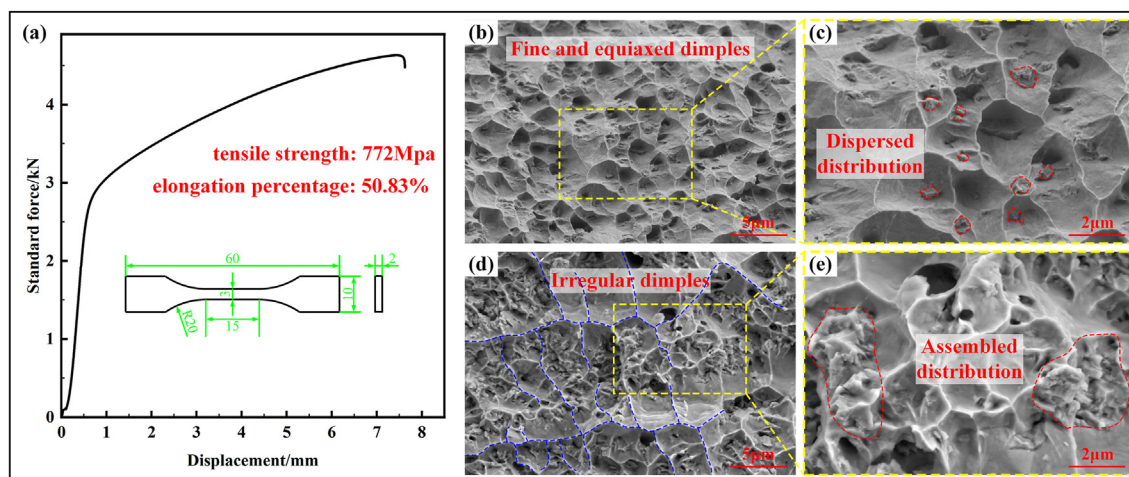


Fig. 14 – Tensile experiments and subsequent fracture analyses. (a) A schematic diagram for tensile sample and resultant curve; (b) and (c) were frequently encountered fracture surfaces; (c) and (e) were enlarged regions in (b) and (c), respectively.

Second, consider the poly-phase eutectic growth. During the formation of γ /Laves constituents, we consider cooling rate ($G \cdot R$) determining not only the length of eutectic reaction zone but also the Nb diffusion for γ /Laves constituent formation. As can be inferred from Table 5, $G \cdot R$ presented an increasing trend from the planar zone to the weld center, thus resulting in a gradually decreased both reaction time and driving force for γ /Laves constituent. Therefore, the fraction of γ /Laves constituents displayed a decreasing trend from the beginning of cellular zone to the ending of dendritic zone, as shown in Figs. 3 and 6. The exceptional situation in the planar zone was mainly caused by relatively limited available locations for γ /Laves constituents under planar S/L. Fig. 13 was partially reprinted from the previous work [39]. The results indicated that even a multi-component melt was capable of presenting a columnar to equiaxed transformation.

Violent and irregular behavior immediately near to the keyhole was an additional factor affecting the solidification process [40]. Turbulence associated with the keyhole is primarily attributed to the ejection of a considerable amount of metal vapor. Therefore, the planar zone was easy to induce an unstable growth condition for solidification. Therefore, not all regions near FL exhibited a typically planar growth. In particular, at the top positions, the additional presence of Marangoni force induced an unstable solid/liquid interface, resulting in a narrowing of the planar zone and a widening and more random orientation in the dendritic zone, as shown in Fig. 2c. Moreover, planar and cellular growth was observed in a relatively short distance from the eruption point, probably inducing the interfacial instability and resultantly finer structures, as shown in Fig. 12. Therefore, eruption induced turbulence was not obvious at the rear of the pool. Consequently, favorable conditions for ordered dendritic existed in the weld center. This also interpreted the generation of fine γ trunks at planar zone with low supercooling and cooling rate while coarse γ trunks at dendritic zone with high supercooling and cooling rate.

4. Tensile performance

Tensile examination was conducted to investigate the variation of microstructure and microtexture generated during the solidification of laser induced weld pool. The schematic diagram of the tensile sample and corresponding tensile curve of standard force vs. displacement were displayed in Fig. 14a. The entire curve presented a naturally ductile fracture feature with a tensile strength of 772 MPa and an elongation percentage of 50.83%. In addition, the fracture site was located near the FL in both top and bottom surface, indicating a relatively higher deformation and fracture resistance in BM and dendritic zone than that in planar and cellular zones.

Secondary electron microscopy was employed to clarify γ and γ /Laves constituents on the fracture surface, as shown in Fig. 14b-e. Two types of fracture surfaces were displayed considering the dimple morphology and Laves distribution. One type (Fig. 14b and c) exhibited a uniformly fine and equiaxed form of dimples whose depressions and ridges were trapped with angular and dispersed particles. Such appearance accorded with the distribution of γ /Laves constituents in the planar zone (Figs. 3, 6 and 13). The other type (Fig. 14d and e) exhibited irregular dimple shape and assembled distribution of Laves phase particles. Note that the ridges exhibited a “dendrite” form, and these particles were primarily found in the “interdendritic” regions. Such phenomena suggested that the γ /Laves constituents acted as initial points of micro cracks or voids and γ trunks were the last-to-fracture positions. The former also corresponded to the GND density in Fig. 5b. As studied, planar and cellular zone presented relatively high SF (Fig. 9) and still possessed medium and large grains though with a relatively low grain size of average value (Fig. 11). Therefore, high SF, relatively large grain size and high fraction of γ /Laves constituents together led to the resultant easy-to-fracture positions in planar and cellular zones of the GH909 laser weldment.

5. Conclusion

In this paper, we described the successful laser welding of a 2 mm GH909 superalloy using parameters of 1.5 kW and 1.5 m/min. Solidification features of the GH909 superalloy from FL to weld center are individually illustrated, including BM, planar, cellular and dendritic zones. Main conclusions are as follows.

- (1) Laser weldment of GH909 created a typical “Y” type seam in which the top and bottom regions expanded in width due to Marangoni flow of the weld pool and eruption of metal vapor. The resultant convection accelerated the unstable solid/liquid interface, resulting in a narrow planar zone and a broad dendritic zone. Apparent planar-cellular-dendritic growth modes were observed only near the vertical and straight fusion line at the mid-penetration.
- (2) Chemically, the partition of Nb, Ti and Si to inter-dendritic regions during the primary growth of γ sub-grains led to subsequent formation of γ /Laves constituents. Texturally, eutectic γ possessed S–N crystal relationship with Laves while retaining the same orientation as the surrounding matrix γ , thereby verifying the existence of a eutectic reaction between γ and Laves. The Nb solution and irregularly shaped Laves resulted in considerable residual strain accommodation in γ /Laves constituents.
- (3) Fiber texture along RD exhibited substantial transformation in γ grains. The texture was altered from $[1\ 5\ 5]_{\text{sample}}$ in BM to a mixture of $[0\ 0\ 1]_{\text{sample}}$ + $[0\ 2\ 5]_{\text{sample}}$ in the planar zone, and subsequently $[0\ 0\ 1]_{\text{sample}}$ in the cellular zone with a gradual increase in pole density. The dendritic zone ultimately presented a concentration at $[0\ 0\ 1]_{\text{sample}}$ with the highest pole density. Furthermore, a relatively high Schmidt factor in the planar and cellular zones and a gradual increase of γ grain size from BM to dendritic zone were observed.
- (4) It was determined that solidification rate was gradually increased while temperature gradient was continuously decreased from fusion line to the weld center. Such variations enhanced the supercooling degree, therefore increasing the primary spacing of γ trunks and γ grain size from planar zone to the dendritic. In addition, such variation also inhibited the formation of γ /Laves constituents. High Schmidt factor, relatively large grain size and high fraction of γ /Laves constituents jointly led to the failure in planar and cellular zones of the GH909 laser weldment.

Declaration of Competing Interest

The authors declare no conflicts of interests.

Acknowledgments

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